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Risposta non ancora data

Punteggio max.: 1,00

Which of the following is not a property of a *metric* distance function

Scegli un'alternativa:

- a. Symmetry
- b. Triangle inequality
-   c. Boundedness
- d. Positive definiteness

[Annulla la scelta](#)Domanda **2**

Risposta non ancora data

Punteggio max.: 1,00

Which of the following *is not* an objective of feature selection

Scegli un'alternativa:

- a. Avoid the *curse of dimensionality*
- b. Reduce the effect of noise
-   c. Select the features with higher range, which have more influence on the computations
- d. Reduce time and memory complexity of the mining algorithms

[Annulla la scelta](#)

Domanda **3**

Risposta non ancora data

Punteggio max.: 1,00

What is the *single linkage*?

Scegli un'alternativa:

- ☐ a. A method to compute the separation of the objects inside a cluster
- ☐ b. A method to compute the distance between two objects, it can be used in hierarchical clustering
- ☒ c. A method to compute the distance between two sets of items, it can be used in hierarchical clustering
- ☐ d. A method to compute the distance between two classes, it can be used in decision trees

[Annulla la scelta](#)

Domanda **4**

Risposta non ancora data

Punteggio max.: 1,00

Which of the following measure can be used as an alternative to the *Information Gain*?

Scegli un'alternativa:

- ☐ a. Rand Index
- ☐ b. Silhouette Index
- ☐ c. Jaccard Index
- ☒ d. Gini Index

[Annulla la scelta](#)

Domanda **5**

Risposta non ancora data

Punteggio max.: 1,00

What is the *Gini Index*?

Scegli un'alternativa:

- ☐ a. A measure of the *entropy* of a dataset
- ☒ b. An impurity measure of a dataset alternative to the *Information Gain* and to the *Misclassification Index*
- ☐ c. An impurity measure of a dataset alternative to *overfitting* and *underfitting*
- ☐ d. An accuracy measure of a dataset alternative to the *Information Gain* and to the *Misclassification Index*

[Annulla la scelta](#)Domanda **6**

Risposta non ancora data

Punteggio max.: 1,00

A Decision Tree is...

Scegli un'alternativa:

- ☐ a. A tree-structured plan of tests on multiple attributes to forecast the target
- ☐ b. A tree-structured plan of tests on single attributes to obtain the maximum purity of a node
- ☐ c. A tree-structured plan of tests on single attributes to forecast the cluster
- ☒ d. A tree-structured plan of tests on single attributes to forecast the target

[Annulla la scelta](#)Domanda **7**

Risposta non ancora data

Punteggio max.: 1,00

When training a neural network, what is the *learning rate*?

- ☐ a. The ratio between the size of the hidden layer and the input layer of the network
- ☒ b. A multiplying factor of the correction to be applied to the connection weights
- ☐ c. The slope of the activation function in a specific node
- ☐ d. The speed of convergence to a stable solution during the learning process

[Annulla la scelta](#)

Domanda 8

Risposta non ancora data

Punteggio max.: 1,00

Which of the statements below about *Hierarchical Agglomerative Clustering* is true?

- a. Is very efficient, also with large datasets
- b. Is based on a well founded statistical model
-   c. Requires the definition of *distance between sets of objects*
- d. Requires the definition of *Inertia* of clusters

[Annulla la scelta](#)

Domanda 9

Risposta non ancora data

Punteggio max.: 1,00

What does K-means try to minimise?

Scegli un'alternativa:

- ☐ a. The *separation*, that is the sum of the squared distances of each cluster centroid with respect to the global centroid of the dataset
- ☐ b. The *distortion*, that is the sum of the squared distances of each point with respect to the points of the other clusters
- ☐ c. The *separation*, that is the sum of the squared distances of each point with respect to its centroid
-   d. The *distortion*, that is the sum of the squared distances of each point with respect to its centroid

[Annulla la scelta](#)

Domanda **10**

Risposta non ancora data

Punteggio max.: 1,00

Which of the following characteristic of data can reduce the effectiveness of DBSCAN?

Scegli un'alternativa:

- ☒ a. All the variables are the same range of values
- ☒ b. Presence of clusters with different densities
- ☐ c. Presence of outliers
- ☐ d. Clusters have concavities

[Annulla la scelta](#)

Domanda **11**

Risposta non ancora data

Punteggio max.: 1,00

Which of the following statements regarding the discovery of association rules is true? (One or more)

Scegli una o più alternative:

- ☒ ☒ a. The confidence of a rule can be computed starting from the supports of itemsets
- ☒ b. The support of a rule can be computed given the confidence of the rule
- ☒ ☒ c. The support of an itemset is anti-monotonic with respect to the composition of the itemset
- ☐ d. The confidence of an itemset is anti-monotonic with respect to the composition of the itemset

Domanda **12**

Risposta non ancora data

Punteggio max.: 1,00

How does *pruning* work when generating frequent itemsets?

SUPERSER
^
SET ^
SUBSET

✓ **Scegli un'alternativa:**

- ☒ a. If an itemset is not frequent, then none of its supersets can be frequent, therefore the frequencies of the supersets are not evaluated
- b. If an itemset is not frequent, then none of its subsets can be frequent, therefore the frequencies of the subsets are not evaluated
- c. If an itemset is frequent, then none of its subsets can be frequent, therefore the frequencies of the subsets are not evaluated
- d. If an itemset is frequent, then none of its supersets can be frequent, therefore the frequencies of the supersets are not evaluated

[Annulla la scelta](#)

Domanda **13**

Risposta non ancora data

Punteggio max.: 1,00

In *feature selection*, what is the Principal Component Analysis?

Vai a...

- ☒ a. A mathematical technique used to transform a set of numeric attributes into a smaller set of numeric attributes which capture most of the variability in data
- b. A mathematical technique used to transform non numeric attributes into numeric attributes
- c. A heuristic technique used to find a subset of the attributes which produces the same classifier
- d. A mathematical technique used to find the principal attributes which determine the classification process

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[Annulla la scelta](#)

Domanda **14**

Risposta non ancora data

Punteggio max.: 1,00

What is the difference between classification and regression?

- a. Classification can have a numeric or categorical target, while regression has always a categorical target
- b. Classification is a supervised activity, while regression is unsupervised
- c. Classification can make errors, while regression is always exact
- ☒ d. Classification has a categorical target, while regression has a numeric target

[Annulla la scelta](#)Domanda **15**

Risposta non ancora data

Punteggio max.: 1,00

Which of the following *is not* a strength point of *Dbscan* with respect to *K-means*

Scegli un'alternativa:

- ☒ a. The *effectiveness* even if there are clusters with non-convex shape
- ☒ b. The efficiency even in large datasets
- c. The *robustness* with respect to the number of attributes
- d. The *effectiveness*, even in presence of *noise*

[Annulla la scelta](#)